

WASP-12b

R-0317

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_231231 · created 2026-07-18T00:53:56+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460309.8686 +/- 0.00082 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1527 +/- 0.0076
Transit depth (Rp/Rs)^2	2.33 +/- 0.23 [%]
Orbital Inclination (inc)	77.17 +/- 1.23
Airmass coefficient 1 (a1)	1.142 +/- 0.012
Airmass coefficient 2 (a2)	-0.1022 +/- 0.0069
Scatter in the residuals of the lightcurve fit is	1.06 %
Best Comparison Star	None
Optimal Aperture	3.99
Optimal Annulus	3.99
Transit Duration (day)	0.1112 +/- 0.0053

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1527 $\pm$ 0.0076 vs 0.1178 (deviation 3.76 σ)
Duration fit vs published	0.1112 d vs 0.125 d (deviation 2.13 σ)
Mid-time O-C	-3.6 min
Depth SNR	16.5
Residual scatter	1.06 %

Light curve

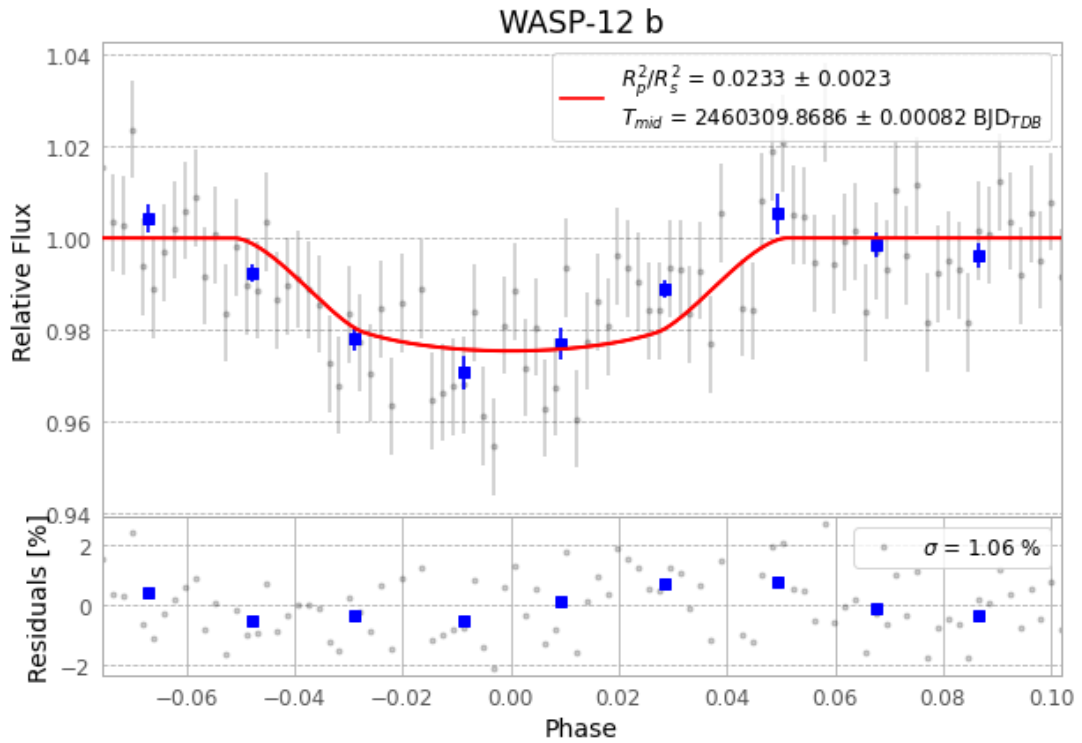


Figure 1 — FinalLightCurve_WASP-12 b_2023-12-30.png (SHA-256 a75e6414e12e483f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [340, 266], 2 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 412111ca5a321629...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1bbef5c

Data provenance

94 FITS frames (62 MB), WASP-12231231064215.FITS → WASP-12231231112115.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-231231.log (34eaa1c1804a...), FinalLightCurve_WASP-12 b_2023-12-30.png (a75e6414e12e...), FinalLightCurve_WASP-12 b_2023-12-30.csv (0f20d1adbb60...)

Compute resources

Wall time 115.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-18 00:53Z.